

100 BEST KNOWN SQUARE PACKINGS

The Squares Project · github.com/jlevy/squares



The square packing problem asks for the side $s(n)$ of the smallest square that can hold n unit squares, where the squares are free to rotate but cannot overlap

- proved optimal (45 of 100)
 - exact value known (97 of 100)
 - only known numerically (3 of 100)
 - rigid (14 of 100)
- recent result, since August, 2026 (81 of 100)
- colors indicate distinct tilt angles
- shade indicates number of full-side contacts
- deg is the algebraic degree of that side length

Best packings due to Göbel, Trump, Wainwright, Härmäläinen, Stenlund, Friedman, Bidwell, Cantrell, Morandi, Ellsworth, Hajba, Schadt, Chaoweerasarit, Couzo, Daniel, de Winter, Rehwaltd, Gupta, ry-xu, Chang, DeVincentis, hmbelvedere

Lower bounds due to Göbel, Stromquist, Kearney, Shiu, Bentz, Daniel, Vlasenko, Guzhou0806, Hosono, Ahmed, Karakuş, Ryu

Optimality proofs due to Göbel, Stromquist, Friedman, El Moumni, Kearney, Shiu, Bentz, Vlasenko, Levy, Daniel, Hosono, Ahmed, Karakuş; formalization contributions by Gurevitch, Guzhou0806; verification infrastructure and execution by EvolvingPrograms, Julian-JJ, Lewis